

1.1 The main hydrological characteristics and processes which operate in rivers and drainage basins

It is crucial to understand what each command word means, so you know what a question is asking you to do. Look at the questions below to see the difference between 'describe' and 'explain' questions. While these questions appear similar at first, the command words demand entirely different responses. Recognising this distinction is an essential skill to develop, so you can focus your answer correctly.

- 1 Describe how the velocity of a river changes as it flows downhill. [2]
- 2 Explain why the velocity of a river changes as it flows downhill. [4]

You will have the opportunity to practise more questions with different command words in each topic of this book.

REFLECTION

Do you understand the difference between 'describe' and 'explain' questions? Seeing them side by side, as with Questions 1 and 2, makes you consider the difference between the two. In this example, 'describe' means say **how** the velocity changes, and 'explain' means say **why** the velocity changes. Creating your own 'describe' and 'explain' questions can help you to recognise the difference.

- 3 Figure 1.1 shows the water cycle, or hydrological cycle. Identify the key terms that correctly complete Boxes A, B and C.

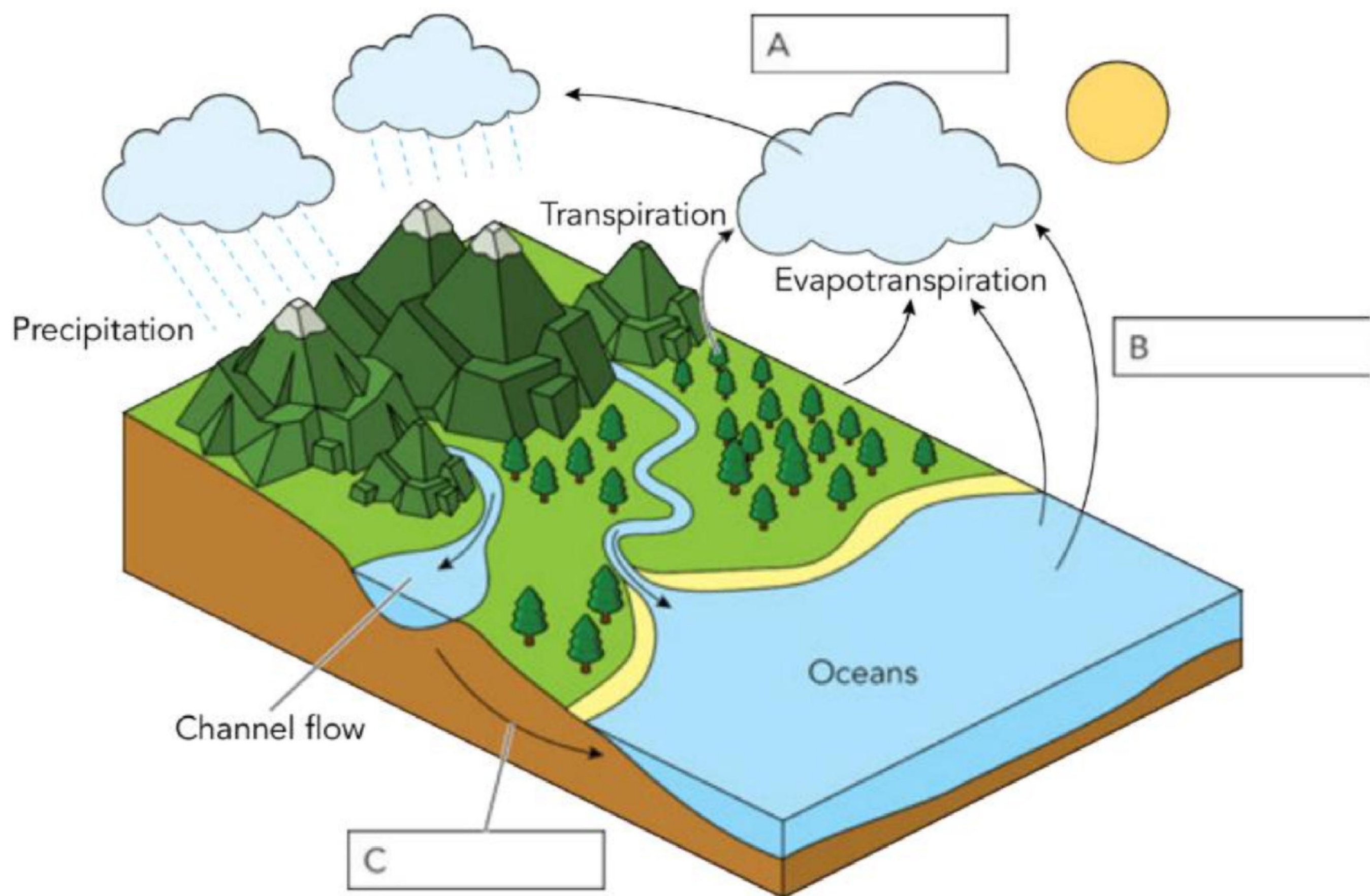


Figure 1.1: The water cycle

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- 4 Copy the following text and complete the gaps using the words below from [Topic 1](#).

watershed tributary upland confluence

A drainage basin is an area of land drained by a river and its tributaries. It moves water from _____ areas towards the sea. The _____ is the boundary between each drainage basin. A _____ is the meeting point between two rivers of similar size, and a _____ is a smaller river or stream that feeds into the main river channel.

- 5 There are three main river processes: erosion, transportation and deposition. Sort the key terms from Table 1.1 into two categories: erosion processes and transportation processes.

traction
hydraulic action
saltation
abrasion
attrition
suspension

Table 1.1: River processes

UNDERSTAND THESE TERMS

- Abrasion
- Attrition
- Drainage basin
- Hydraulic action
- Saltation
- Suspension
- Traction
- Wetted perimeter

- 6 Which processes of erosion and transportation are missing from Table 1.1?

Questions with the command word 'explain' require you to show that you understand how or why something occurs. These questions test

your ability to connect ideas and apply your knowledge effectively. See if you can answer these 'explain' questions.

- ~~7~~ Explain how erosion, transportation and deposition work together in a river. [4]
- ~~8~~ Explain why vertical erosion is more likely to take place in the upper course of a river. [4]
- ~~9~~ Figure 1.2 shows a lowland river. Explain why rivers in the lower course have a larger wetted perimeter. [2]



Figure 1.2: A lowland river

- ~~10~~ Explain why alluvium (deposited material) is often found in the middle and lower courses of a river. [4]